

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: **BIT101R01**

Course: **BIOLOGY FOR ENGINEERS**

QP No. :U011R-1

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. The patterned surface roughness of shark skin prevents bacterial growth and biofilm formation. Apply this natural phenomenon for biomimicry applications in engineering?
2. Mr. Kumar was complaining of abdominal gas, severe stomach pain and diarrhoea. His last meal was enriched with dairy products. What could be the reason for his discomfort? What would be your recommendations to avoid such conditions?
3. Airport security personnel employ methods to protect passengers, staff, and aircraft from malicious harm, crime, terrorism, and other threats. A similar function is also performed in our biological system that selectively allows the passage of certain molecules. Identify it and draw its structure.
4. Storing military information is one of the important issues of this century. Heavy reliance on computers for communications, vehicle control, surveillance, and signal processing make it imperative for the forces to keep the data secure. Information needs to be stored for many years which can provide an idea for the upcoming military personnel on how to execute plan and handle unexpected problems during tough situations. What could be your

technical advice on solving the problem of storing military secret data for a long period of time?

- 5/ An automobile company is in pursuit of designing the best Alpine race car. Though, there are several hundreds of factors including the material used for the car body, height, wheel width, wheelbase design, overall design, the engine to consider, it may not be possible for the company to come up with a race car design quickly. Suggest a bio-inspired technology which would save their time and energy, and brief about the design step process.
- 6/ The use of human embryonic stem cells raises ethical issues world-wide. Comment on the ethical dilemma and suggest alternate sources for stem cell research and therapy.
- 7/ Imagine you are a scientist assigned with a job to reduce the years of drug testing, since the drugs available in the nearby pharmacy is an outcome of several years of research. Mention the latest available technology to reduce drug testing and add its advantages.
- 8/ Engineers have developed machines that can interact and work together with biological systems. Give two such examples of machines/technology and explain briefly the components that interact with each other in these set-ups.
- 9/ What are the components of 'bioink' used for printing human tissues/organs?
- 10/ You plan to work in a microbiology lab to study the growth of pathogenic E. coli strain, which is expected to cause health-hazard to humans. Mention the precautions that you should follow during these experiments.

PART - B

Answer any four questions

4 x 15 = 60 Marks

11. a) Specially formulated high-performance color pigments and binder technology, which offer excellent resistance to ultraviolet radiation, dust and microbial colonization. Highlight the bioinspired technology behind this concept that ensures dark spots and microbes stay away (5)
- b) The architects and builders are seeking new and improved ways to cool buildings without using conventional air conditioning. An engineer has got an important clue from nature that inspired him to design and make buildings, which could easily eliminate the huge heating and cooling plants and thereby

reduce maintenance cost. Which bioinspired principle could be involved behind this successful idea? (5)

c) The woodpecker's head is an excellent example of a shock absorption system. It does not suffer injury even while hitting a tree with an acceleration much higher than the impact that causes traumatic brain injury in humans. The reasons for this resilience are thought to be (i) a sharp stoutly pointed elastic beak, (ii) cerebrospinal fluid for brain cushioning and, (iii) spongy bone material behind the skull. Propose an engineering application/technology that uses all these features. (5)

12. a) Imagine a crack has appeared in the walls of the building and must repair on its own now. Explain the mechanism of action and the principle involved. (10)

b) Mention the at-least two industrial applications of the following Biomolecules, (i) Carbohydrate, and (ii) Lipid. (2 ½ x 2 = 5 marks)

13. An adventurous person has planned for trekking. He can carry a maximum of 35kg materials with him in his survival kit. Below are the items with specific weight &

S.No	Item	Weight (kg)	Survival point
1.	Glucose	20	17
2.	Rope	10	10
3.	Sleeping bag	15	15
4.	Emergency light	5	8
5.	Knife	2	12
6.	Medicines	10	6

How will you address this problem in carrying the weight of 35kg with maximum survival points (15)

14. Blood purification has been applied conventionally in the management of patients with multiple organ failure (MOF). Recent advances in medical engineering made it possible to perform such blood purifications continuously (i. e., 24 hours a day, 7 days a week if necessary) even in critically ill patients. The continuous blood purification can remove or decrease the blood levels of humoral mediators, including proinflammatory cytokines, and can improve tissue oxygenation, especially oxygen consumption (VO₂) among critically

ill patients. Discuss on the engineering application that can be used for blood purification (15)

15. Natural neurons receive signals through synapses located on the dendrites, or membrane of the neuron. When the signals received are strong enough (surpass a certain threshold), the neuron is activated and emits a signal through the axon. This signal might be sent to another synapse and might activate other neurons. Based on the inspiration from the architecture of the human brain and its biological neural networks, a technology has been developed and utilized for various applications. Discuss on it. (15)

16. a) A biosafety level (BSL) is a set of biocontainment precautions required to isolate dangerous biological agents in an enclosed laboratory facility. In the United States, the Centers for Disease Control and Prevention (CDC) have specified these levels. The levels (BSL-1,2,3 and 4) are designated in ascending order, by degree of protection provided to personnel, the environment, and the community. Discuss on the precaution that needs to be taken into consideration while entering into BSL-1,2,3 and 4 (7 ½)

b) Biological Waste Disposal Policy stipulates proper procedures for the collection, decontamination, and disposal of laboratory-generated biohazard waste. How the biowaste needs to be categorized and discarded. (7 ½)

PART – C

1 x 20 = 20 Marks

Answer the following

17. Constructing a multi-storied and secured building requires an integrated management. As an engineering expert, give your suggestion for the following features from the bioinspired concepts you have learned. Also, discuss how you would use these ideas for incorporation of these features in the building. (4 marks each)

i. To fix anti-reflecting glass windows

ii. To remove the malodorous gases in the rooms

iii. Face recognition security system to prevent entry of unauthorized personnel

iv. To construct more stable building in high speed cross winds

v. Rooms maintained at specific temperature and humidity
